

Sooham Rafiz

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Career Break Independent Researcher and Study @ Fudan University

Jul 2025 - Present

- Took career break following significant investment gains, enabling focus on research interests and extended stay in Shanghai, China studying Chinese and Physics. Reached HSK level 5 in Shanghai. 我在复旦学中文和物理, 都是因为好奇。达到 HSK 五级水平。
- Replicated and Extended (Noroozizadeh et. al) paper "Deep sequence models tend to memorize geometrically; it is unclear why." [Blog post](#)
- Practical research into Pathstar optimization with mmap and sorting techniques. [Blog post](#)
- Self-teaching: Quantum Physics, Classical Mechanics.
- Performance optimization projects on cache line investigation, CUDA, kernel optimization. [Blog post on Apple M4 cache hierarchy](#)

Short Story (YC'19) Senior Software Engineer / ML Engineer

Apr 2024 - Jul 2025

- Leveraged open-source vision language model cogVLM-17B + LoRA with propriety data to train, validate, test and deploy fine-tuned fashion cataloging and embedding utilities used throughout company - used for automated cataloging of inventory. F1 score of 0.861 and accuracy of 88% on confusion matrix.
- Designed and implemented a ShipmentEngine utilized by merchandising team to track, bid and purchase 100k+ inbound and outbound shipments between company and vendors.
- Saved 1.8 million USD per annum in freight costs by switching away from USPS to Pitney Bowes E-commerce platform. Spearheaded the technology and engineering for transition for Customers, Customer Support and Warehouse Operations.
- Designed an in-house cycle-counting and supporting data ingestion pipelines to eliminate all inactive warehouse zones and improve inventory count accuracy.
- **Utilized:** *Pytorch, DDP, Postgres, Typescript, React, AWS, Retool, Falcon, Heroku, Python*

Amazon Software Development Engineer II - Payments Team

December 2021 - Nov 2023

- Developed real-time Unpaid Time value algorithm and infrastructure for Germany - moving away from batched 3rd party vendor dependency used previously. Speed up by a factor of 1000. Deployed to 2 million warehouse associates.
- Cross-functionally led the development of a cutting-edge timesheet analysis software for 1k daily workers in Puerto Rico Amazon Logistic Center.
- Developed a new model of computation for missed meal payment, significantly reducing overpayment when warehouse associates miss meal breaks.
- Mentored engineers, wrote a portfolio of team specific onboarding documentation, videos tutorials for future customers and new-hires and operational efficiency.
- **Utilized:** *Python, Java, AWS ECS, AWS DDB, Lambda, CloudWatch, Cloudformation, CDK*

Qualcomm Machine Learning Acceleration Engineer

July 2019 - August 2021

- Built a python compiler to convert JSON denominated computational graphs to standardized ONNX graphs which supported Qualcomm proprietary kernels for Layer fusion, vector precision adjustment.
- Developed an end-to-end CI/CD ML test pipeline and performance dashboard to test model quantization on 100+ vision model families (Inceptionv3, MobileNet, YOLO etc.) the dashboard profiled timing, memory footprint and quantization error across multiple target hardware devices and tensor types (quantized and dequantized).
- Implemented a static quantization library to quantize floating-point neural network inputs and weights before inference for Snapdragon Neural Processor firmware.
- Reduced memory footprint of common operators such as Convolution, Argmax etc compiled binaries.
- **Utilized:** *C, Python, Pytorch, ONNX, Qualcomm Hexagon SDK, MLIR*

Intel Corporation Deep Learning Acceleration Engineer (Intern)

Jun 2018 - Aug 2018

- Reduced OpenVINO (a deep learning framework) compile time for CNN models by upto 24 hours using caching optimizations.
- Reduced compile time for building deep convolutional models in regression tests by caching intermediary C object files, libraries and configurations to a Python server, decreasing test time by 16 hours and improving test repeatability.

- Migrated Intel's deep learning framework used by 50+ developers across backwards incompatible updates to development and production environment, sustaining team productivity.
- **Utilized:** *C++, Python, OpenVINO, OpenCL*

Intel Corporation FPGA Development Engineer (Intern)

Jun 2017 - May 2018

- Created a comprehensive user interface and packaging system using Perl and Tcl, allowing parameter customization, code encryption and code generation. Used in production by multiple customers.
- Built a C++ client and server to query field-programmable gate array (FPGA) registers. This was accomplished by sending Direct Memory Access and Memory Mapped IO requests to FPGA workstations.
- Added FPGA hardware support to Intel's digital signal library by implementing C++ middleware.
- **Utilized:** *C++, Perl, Tcl, Quartus, Verilog*

Education

University of Toronto Honors B.Sc. Computer Science

2014 - 2019

Projects

[Hyperloglog Cardinality Estimator](#)

May 2019

- Compiled C++ HyperLogLog randomized algorithm used for efficient cardinality estimation.

[Image Inpainting C++ Library](#)

June 2016 - Aug 2016

- Authored the only open-source implementation of Criminisi et. al's Inpainting algorithm in C++ as an online reference.

Technologies

Languages: Python, Java, Ruby, SQL, Typescript, React, Node.js

Technologies: AWS - DynamoDB, EC2, ECS, Lambda, EKS, Cloudformation, Git, NoSQL